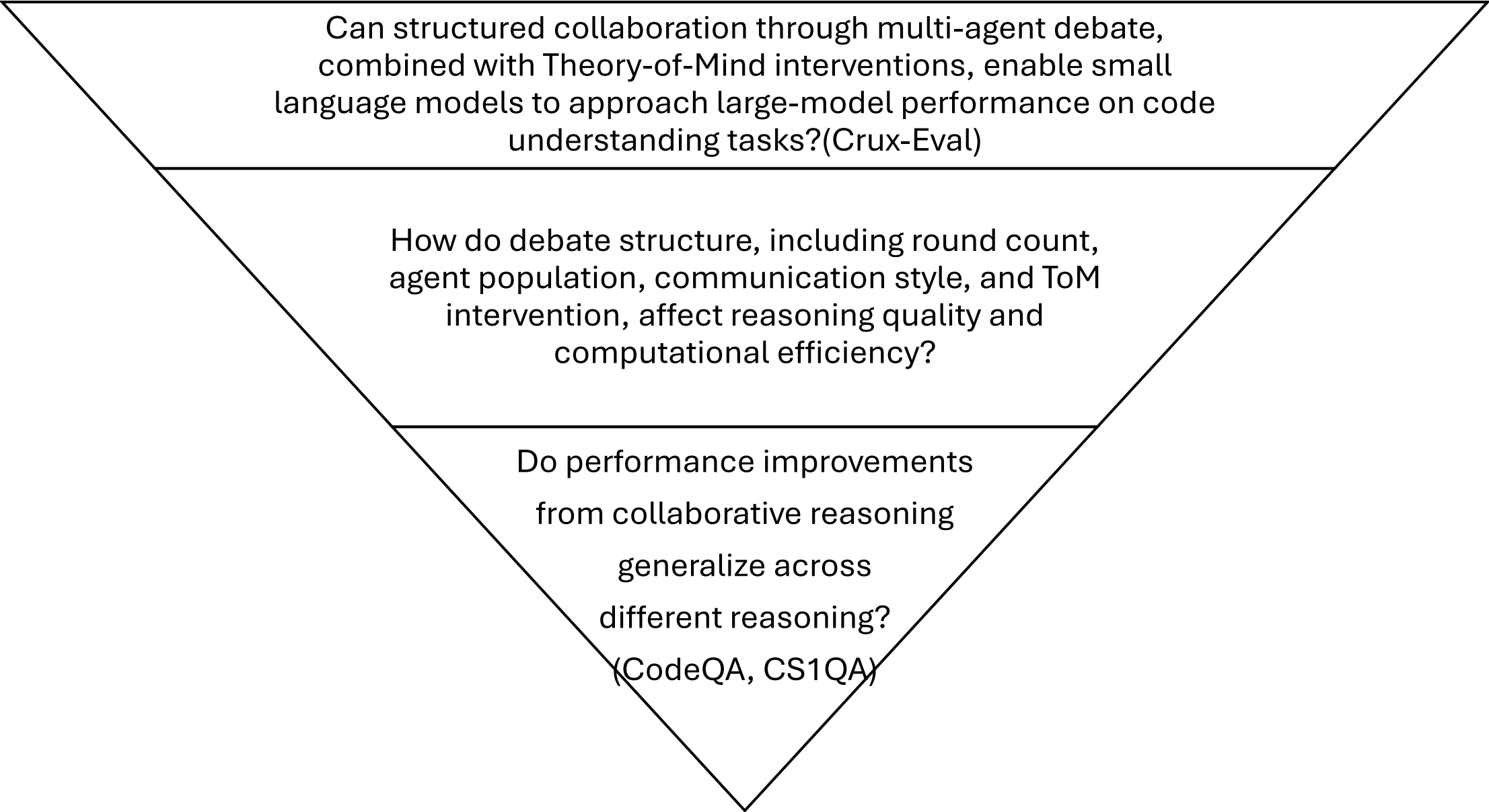


Motivation

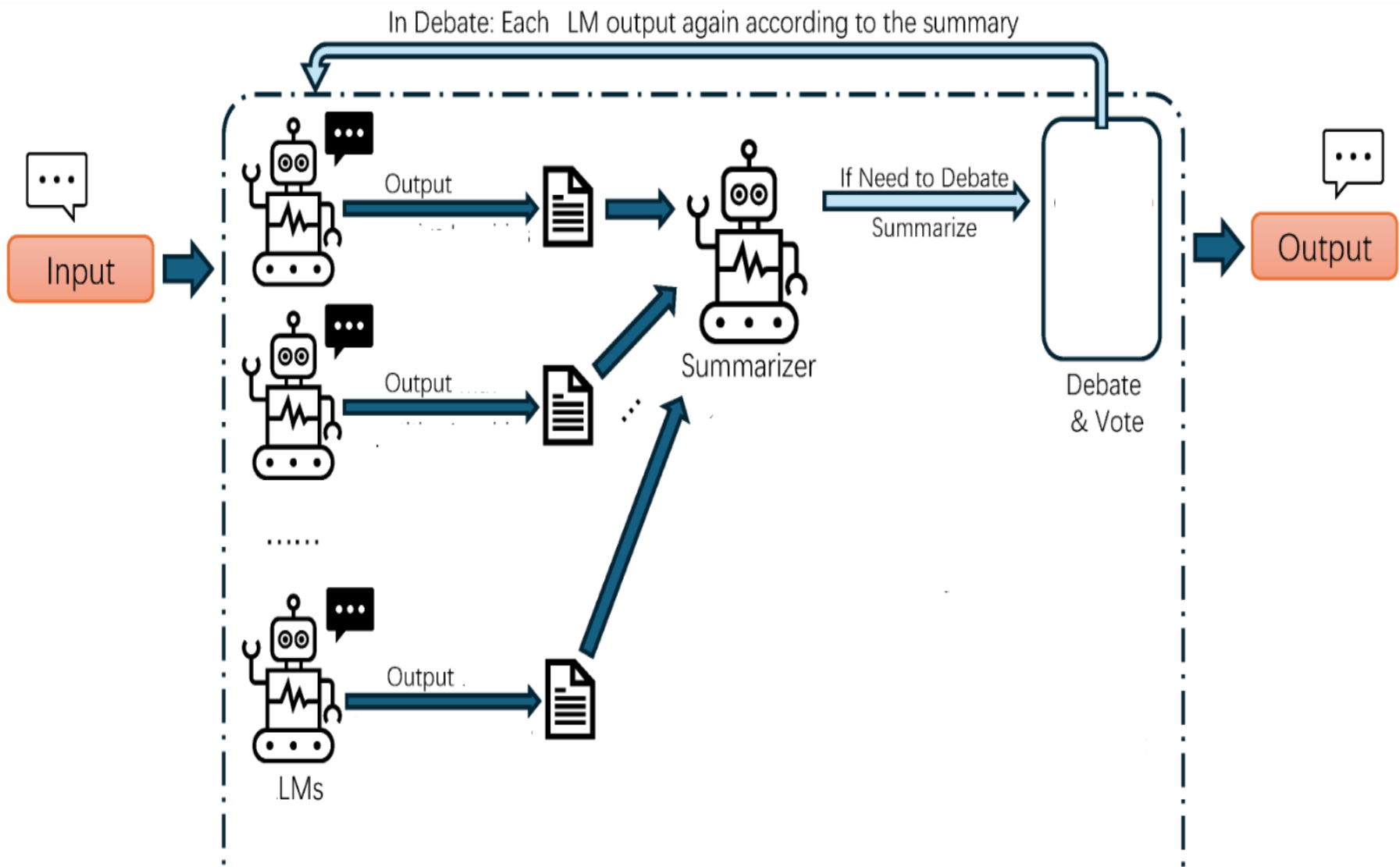
- Problem: Limited code understanding capabilities in small language models as compared to large language models
- Solution: examine whether structured reasoning frameworks can improve small-model code understanding capabilities.
- What are existing solutions: Multi-agents, ToM
- Technique(old), Problem(new)
-



Can structured collaboration through multi-agent debate,
combined with Theory-of-Mind interventions, enable small
language models to approach large-model performance on code
understanding tasks?(Crux-Eval)

How do debate structure, including round count,
agent population, communication style, and ToM
intervention, affect reasoning quality and
computational efficiency?

Do performance improvements
from collaborative reasoning
generalize across
different reasoning?
(CodeQA, CS1QA)



TOM, Post, Inter

Input to Judge:

- Original question
- All debate rounds (Agent 1, 2, 3 arguments across R rounds)
- Evaluation criteria (A/C/R/C rubric)

Judge Prompt:

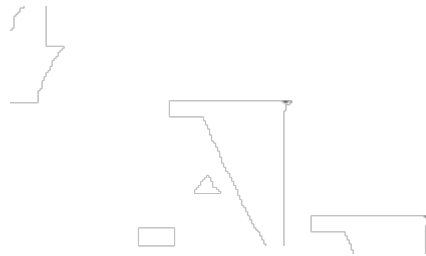
"You are an expert judge reviewing a multi-agent debate about the following code comprehension question:

[QUESTION]

The following agents have debated this question across [R] rounds:

Round 1:

- Agent 1: [response]
- Agent 2: [response]
- Agent 3: [response]



Round 2:

...

Your task:

1. Identify points of consensus among agents
2. Evaluate conflicting viewpoints with evidence
3. Synthesize the most accurate, complete, and clear answer
4. Explicitly state which agent arguments you found most persuasive and why

Provide your final synthesized answer."